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AWS Shared Responsibility Model

The **AWS Shared Responsibility Model** defines how security responsibilities are shared between Amazon Web Services and the customer.

Responsibilities

1. **AWS Responsibility (Security *of* the Cloud)**
   - Physical security of data centers
   - Hardware and infrastructure
   - Networking and virtualization
2. **Customer Responsibility (Security *in* the Cloud)**
   - Data protection
   - Operating system and applications
   - Identity and access management

AWS Identity and Access Management (IAM)

AWS Identity and Access Management (IAM) is a service provided by Amazon Web Services that controls access to AWS resources securely.

Key points:

1. **Users** – Individual accounts for people or applications.
2. **Groups** – Collection of users with similar permissions.
3. **Roles** – Temporary access permissions for services or users.
4. **Policies** – JSON documents that define permissions (allow/deny).
5. **Security** – Supports multi-factor authentication (MFA).
6. **Fine-Grained Access Control** – Users get only required permissions.

Security Credentials

Security Credentials are used to authenticate and authorize users in Amazon Web Services.

Key types:

1. **Root User Credentials** – Email and password used to create AWS account (full access).
2. **IAM User Credentials** – Username and password for accessing AWS services via AWS Identity and Access Management.
3. **Access Keys** – Access Key ID and Secret Access Key used for programmatic access (CLI/API).
4. **Key Pairs** – Public and private keys used to securely connect to Amazon EC2 instances.
5. **Temporary Credentials** – Short-term credentials provided by IAM roles.
6. **Multi-Factor Authentication (MFA)** – Adds extra security layer.

AWS Trusted Advisor

AWS Trusted Advisor is a service provided by Amazon Web Services that gives recommendations to improve AWS usage.
It checks resources in the following areas:

1. **Cost Optimization** – Suggests ways to reduce unnecessary spending.
2. **Performance** – Improves system speed and efficiency.
3. **Security** – Identifies security gaps and risks.
4. **Fault Tolerance** – Ensures high availability and reliability.
5. **Service Limits** – Monitors usage limits of AWS services.

AWS CloudTrail

AWS CloudTrail is a service provided by Amazon Web Services that records and tracks user activity and API calls in AWS.
Key points:

1. **Activity Logging** – Records all actions taken in AWS account.
2. **API Monitoring** – Tracks API calls made by users or services.
3. **Security & Auditing** – Helps detect unauthorized access.
4. **Event History** – Maintains history of events for analysis.
5. **Integration** – Works with services like Amazon CloudWatch.
6. **Compliance** – Helps meet auditing and regulatory requirements.

AWS Config
AWS Config is a service provided by Amazon Web Services that monitors and records configuration changes of AWS resources.
Key points:

1. **Resource Tracking** – Tracks configuration of AWS resources over time.
2. **Configuration History** – Maintains history of changes.
3. **Compliance Monitoring** – Checks whether resources follow rules and policies.
4. **Change Detection** – Detects and alerts when changes occur.
5. **Integration** – Works with Amazon CloudWatch for alerts.
6. **Auditing** – Helps in security analysis and compliance.

AWS Day One Best Practices

AWS Day One Best Practices are guidelines recommended by Amazon Web Services to ensure a secure and efficient cloud setup from the beginning.
Key practices:

1. **Secure Root Account** – Enable MFA and avoid using root user regularly.
2. **Use IAM Users** – Create users with limited permissions using AWS Identity and Access Management.
3. **Enable Billing Alerts** – Monitor costs using Amazon CloudWatch.
4. **Enable Logging** – Use AWS CloudTrail to track activities.
5. **Set Up VPC** – Use Amazon VPC for network isolation.
6. **Backup Data** – Use storage services like S3 for backups.

AWS Security & Compliance Programs

AWS Security & Compliance Programs are policies and certifications provided by Amazon Web Services to ensure data protection and regulatory compliance.

Key points:

1. **Security Standards** – AWS follows global security standards to protect data.
2. **Compliance Certifications** – Includes ISO, SOC, PCI-DSS, etc.
3. **Data Protection** – Encryption, access control, and monitoring.
4. **Shared Responsibility Model** – Security is shared between AWS and customers.
5. **Audit Support** – Helps organizations meet legal and regulatory requirements.
6. **Global Compliance** – Supports multiple industries and regions.

AWS Security Resources

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AWS Security Resources are tools and services provided by Amazon Web Services to help secure cloud environments.

Key resources:

1. **AWS Identity and Access Management (IAM)** – Controls user access and permissions.
2. **Amazon CloudWatch** – Monitors system performance and security events.
3. **AWS CloudTrail** – Records user activities and API calls.
4. **AWS Config** – Tracks configuration changes.
5. **AWS Trusted Advisor** – Provides security and best practice recommendations.
6. **Encryption Services** – Protect data using encryption techniques.

Security Compliance

Security Compliance refers to following rules, standards, and regulations to ensure data protection and secure systems. In cloud computing, it is supported by Amazon Web Services.

Key points:

1. **Regulatory Standards** – Compliance with standards like ISO, SOC, PCI-DSS.
2. **Data Protection** – Ensures confidentiality, integrity, and availability of data.
3. **Policies & Procedures** – Organizations must follow defined security policies.
4. **Auditing** – Regular checks to ensure compliance.
5. **Risk Management** – Identifies and reduces security risks.
6. **Legal Requirements** – Ensures adherence to laws and regulations.

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AWS Well-Architected Framework

The **AWS Well-Architected Framework** is a guideline developed by Amazon Web Services to help build **secure, high-performing, resilient, and efficient cloud architectures**.

Six Pillars

1. **Operational Excellence**
   - Monitor systems and improve processes
   - Automate operations
2. **Security**
   - Protect data, systems, and assets
   - Implement IAM, encryption, and monitoring
3. **Reliability**
   - Ensure system availability
   - Recover quickly from failures
4. **Performance Efficiency**
   - Use resources efficiently
   - Select the right services and configurations
5. **Cost Optimization**
   - Avoid unnecessary expenses
   - Use pay-as-you-go model effectively
6. **Sustainability**
   - Minimize environmental impact
   - Optimize resource usage

Benefits

- Improves system design
- Enhances security and reliability
- Reduces cost and risk
- Provides best practices for cloud architecture

Architectural Pillars

The **Architectural Pillars** are the six key principles of the **AWS Well-Architected Framework** provided by Amazon Web Services for designing efficient cloud systems.

Six Pillars

1. **Operational Excellence** – Efficient system operation and monitoring
2. **Security** – Protection of data and systems
3. **Reliability** – Ability to recover from failures
4. **Performance Efficiency** – Optimal use of resources
5. **Cost Optimization** – Minimizing unnecessary expenses
6. **Sustainability** – Reducing environmental impact

Design Principles

Design Principles are general guidelines provided in the **AWS Well-Architected Framework** by Amazon Web Services to build efficient and secure cloud systems.
Key Design Principles

1. **Stop guessing capacity** – Use scalable resources instead of fixed capacity
2. **Test systems at production scale** – Check performance under real conditions
3. **Automate to make changes easier** – Use automation for deployment and management
4. **Allow for evolutionary architectures** – Design systems that can improve over time
5. **Drive architectures using data** – Make decisions based on monitoring data
6. **Improve through game days** – Simulate failures to improve system reliability

Reliability & High Availability

Reliability & High Availability are key concepts in cloud computing that ensure systems work continuously without failure. These are core principles in the **AWS Well-Architected Framework** by Amazon Web Services.
Key Points

1. **Reliability** – Ability of a system to function correctly over time.
2. **High Availability (HA)** – Ensures system is always accessible with minimal downtime.
3. **Fault Tolerance** – System continues working even if one component fails.
4. **Backup & Recovery** – Data is backed up and can be restored quickly.
5. **Redundancy** – Multiple resources are used to avoid single point of failure.
6. **Load Balancing** – Distributes traffic to maintain system performanc

Business Impact

Business Impact refers to the effect of IT systems or failures on business operations. In cloud computing with Amazon Web Services, it helps organizations understand risks and ensure continuity.
Key Points

1. **Downtime Impact** – Loss of revenue and productivity during system failure.
2. **Customer Experience** – Poor performance affects user satisfaction.
3. **Financial Loss** – System issues can lead to direct and indirect losses.
4. **Reputation Damage** – Frequent failures reduce trust in the business.
5. **Data Loss Risks** – Loss of important data affects operations.
6. **Business Continuity** – Need for backup and recovery strategies.

Example: Data Center → Cloud

Data Center to Cloud Migration means moving IT infrastructure from on-premises data centers to cloud platforms like Amazon Web Services.

Example

- A company stores data and runs applications in a local data center.
- It migrates to cloud services like Amazon EC2 (compute) and Amazon S3 (storage).

Benefits

1. **Reduced Cost** – No need to maintain hardware
2. **Scalability** – Resources can be increased easily
3. **High Availability** – Cloud ensures better uptime
4. **Security** – Advanced cloud security features
5. **Remote Access** – Access from anywhere
6. **Backup & Recovery** – Easier data protection

AWS Billing & Cost Management

AWS Billing and Cost Management is a comprehensive tool by Amazon Web Services that helps users track, analyze, and optimize their cloud spending.

Key Components

1. **Billing Dashboard**
    - Displays real-time cost and usage summary
2. **Cost Explorer**
    - Visual tool to analyze historical and forecasted costs
3. **AWS Budgets**
    - Set custom budgets
    - Receive alerts when limits are exceeded
4. **Cost & Usage Reports (CUR)**
    - Detailed reports of resource usage
    - Helps in cost analysis
5. **Cost Allocation Tags**
    - Assign tags to resources
    - Track spending by project or team
6. **Alerts & Notifications**
    - Integrated with Amazon CloudWatch

Benefits

7. **Cost Control** – Prevents overspending
8. **Transparency** – Clear visibility of expenses
9. **Optimization** – Helps reduce unnecessary costs